

CoSA: An Accelerated ISTA Algorithm for Dictionaries Based on Translated Waveforms

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Abstract

The use of compressive sensing techniques has become predominant in signal processing applications when dealing with sparse signals. Theoretical results on these tools involve drastic conditions on the dictionaries used, which are usually not met in practice. Nevertheless, it has been recently shown that even in the case of very correlated dictionaries, such as dictionaries based on translated waveforms, information can be retrieved from the observations with good accuracy. However, in numerous fields the signals that we have to process are extremely large, making the direct use of standard algorithms impossible. This paper presents a fast version of the Iterative Shrinkage-Thresholding Algorithm (ISTA) for sparse reconstruction based on dictionaries built from translated waveforms. By rearranging the columns of the dictionary, we show that matrix-vector multiplications can be alternatively expressed as convolutions, which can be performed efficiently using standard Fast Fourier Transforms (FFTs). Results on synthetic data validate the proposed convolutional ISTA (CoSA) method. Furthermore, we also show that CoSA is able to attain good results in

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a real-world application, where ISTA cannot be applied directly: spike localization and activity estimation in a large spectroscopic signal.

Keywords: Compressive sensing; Sparse reconstruction; Iterative Shrinkage-Thresholding Algorithm (ISTA); FFT.

1. Introduction

Since the development of compressive sensing theory and applications, the use of convex optimization techniques has become more and more popular. Among other techniques, methods based on thresholded gradient [1] converge to a sparse solution quickly [2, 3] and have been widely used in signal processing applications [4]. Indeed, in many applications (such as spectroscopy [5], ECG signal processing [6], network traffic analysis [7] or image processing [8]) we are interested in the sparse reconstruction of a recorded signal. For example, in the field of nuclear spectroscopy we wish to retrieve arrival times and energies of recorded electrical pulses, whereas in ECG signal processing we are typically interested in the beat-to-beat rate. In these cases, it has been shown empirically (see e.g. [5, 9, 10, 11, 12, 13]) that we can rely on correlated dictionaries composed of multiple translated waveforms and obtain good results. Furthermore, recent developments in the field include theoretical investigations on correlated dictionaries [14].

The main shortcoming for the use of sparse reconstruction methods in all of these frameworks is the algorithmic complexity, stemming from the fact that we often have to process very large data sets. This leads to the use of extremely large dictionaries, and time consuming matrix computations. So far, the standard solution in the literature involves cutting the large signals into small chunks, which are more convenient to proceed. However, this comes at the cost of the precision of the reconstruction, particularly at the tails of the chunks. The objective of this paper is to provide a fast algorithm to deal with large signals. We first recall the mathematical model representing the signal at hand, as well as the principles of the Iterative Shrinkage-Thresholding Algorithm

(ISTA). We then show that the most time-consuming operations of ISTA can be reduced to convolutions and correlations, which can be computed quickly and efficiently using zero-padded Fast Fourier Transform (FFT) multiplications. This can be related to [8], in which a proximal method was simplified by replacing a matrix inversion by a FFT. However, their algorithm still involves a matrix-vector multiplication, and it thus needs to store the dictionary in RAM, making it ineffective for large signals from spectroscopy and ECG recordings. Finally, we illustrate the speed up provided by the proposed Convolutional Sparse Approximation algorithm (CoSA), both in synthetic signals and a very large signal stemming from real-world spectroscopic data.

2. Fast Algorithm for Sparse Reconstruction with Dictionaries Based on Translated Waveforms

2.1. Model Description

We briefly recall the standard model used in [5, 9, 10]. Let us assume that we observe N points of a sampled signal, y_n , consisting of electrical pulses of different shapes and amplitudes,

$$y_n = \sum_{k=-\infty}^{\infty} E_k \Phi_k(t_n - T_k) + \varepsilon_n, \quad n = 0, \dots, N-1,$$

where T_k denotes the starting time of the k -th electrical pulse; E_k its amplitude; Φ_k the associated, unknown, pulse shape; and ε_n an additional noise, assumed to be Gaussian and with known variance throughout the paper. Eq. (??) is known in the literature as the *shot-noise model*, and is commonly used in the fields of nuclear spectroscopy and biomedical engineering, among others. In these applications, neither Φ_k nor T_k and E_k are known. However, if the dynamics of Φ_k is known, then we can approximate it by means of a dictionary of known waveforms. More precisely, we define a set of p candidate waveforms to characterize the signal dynamics as $\Gamma_s, s = 1 \dots p$, all of them with a finite support of M sampling points along the time scale. The shapes Γ_s are used to estimate the local behavior of the signal around all the sampling points. That is, we define

the following $N \times p$ matrices $\mathbf{A}_k, k = 0 \dots N - M - 1$ whose columns entries are equal to $\Gamma_s(t_i - t_k)$ for $i = k \dots k + M - 1$ and 0 otherwise, as displayed in Figure 1. Note that $\mathbf{A}_k$ can be interpreted as a sub-dictionary containing all the different electrical pulses starting at the same sampling time t_k . Along the paper, it is assumed that $N \gg M$, in order to include shapes entirely at least once. Note that this usually happens in the sparse reconstruction framework, thus this additional assumption implies little loss of generality. A global dictionary $\mathbf{A}$ of size $N \times (N - M)p$, including all possible pulses occurring at different times, can then be defined by regrouping the $\mathbf{A}_k$'s altogether:

$$\mathbf{A} = [\mathbf{A}_0 \quad \mathbf{A}_1 \quad \dots \quad \mathbf{A}_{N-M-1}] \quad .$$

Given any vector $\boldsymbol{\beta}$ of size $(N - M)p$, it can thus be naturally decomposed into blocks of size p : $\boldsymbol{\beta} = [\boldsymbol{\beta}_0^T, \dots, \boldsymbol{\beta}_{N-M-1}^T]^T$, with $\boldsymbol{\beta}_i \in \mathbb{R}^p$ for all i . Thus, the model investigated in the rest of the paper is

$$\mathbf{y} = \mathbf{A}\boldsymbol{\beta} + \boldsymbol{\varepsilon} + \boldsymbol{\delta} = \sum_{k=0}^{N-M-1} \mathbf{A}_k \boldsymbol{\beta}_k + \boldsymbol{\varepsilon} + \boldsymbol{\delta},$$

where $\mathbf{y} = [y_0, y_1, \dots, y_{N-1}]^T$ and $\boldsymbol{\varepsilon} = [\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{N-1}]^T$ represent, respectively, the recorded signal and the additive noise; $\boldsymbol{\beta}$ is a sparse vector whose non-null entries are related to photon arrival times in spectroscopy and heartbeats in electrocardiography; and $\boldsymbol{\delta} = [\delta_0, \delta_1, \dots, \delta_{N-1}]^T$ is a vector quantifying the discretization error made between (??) and (??). We assume that the sampling frequency is high enough so that the influence of $\boldsymbol{\delta}$ is negligible, and thus it can be safely ignored in the following.

In numerous signal processing applications, we are interested in deducing some information from a sparse reconstruction of $\mathbf{y}$. For example, in the field of nuclear spectroscopy, the times T_n form a sample path of a Poisson process, and it is of interest to estimate its activity [5]. A similar problem appears in the study of intracardiac ECGs, where we are interested in finding the position of electrical pulses in each lead of the recording [9]. In previous contributions, (??) was investigated using standard sparse reconstruction techniques, such as

the Least Absolute Shrinkage and Selection Operator (LASSO)[15]. Namely, in the applications considered in [5, 9] the crucial step for signal characterization required solving

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{A}\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1,$$

where λ is a parameter defining the trade-off between the sparsity of $\boldsymbol{\beta}$ and the precision of the estimation.

Unfortunately, for some of the most commonly used methods, such as the standard LASSO or the Least-Angle Regression (LARS) algorithm [16], solving (??) implies building the full matrix $\mathbf{A}$ and storing it in memory. For example, ISTA (and its speeded up version FISTA, see [3]) relies on iterations of a soft-thresholding gradient. More precisely, we compute until convergence

$$\boldsymbol{\beta}^{(k+1)} = T(\boldsymbol{\beta}^{(k)} + 2\mu_k \mathbf{A}^T (\mathbf{y} - \mathbf{A}\boldsymbol{\beta}^{(k)}), \lambda\mu_k),$$

where $T(\mathbf{x}, \alpha) = (|\mathbf{x}| - \alpha)_+ \text{sign}(\mathbf{x}) = \left[\max\{|x_i| - \alpha, 0\} \text{sign}(x_i) \right]_{1 \leq i \leq (N-M)p}$, λ is the sparsity parameter used in the LASSO; and μ_k is the step used in the gradient. In this procedure, the bottleneck in the implementation of (??) is the multiplications by $\mathbf{A}$ and by $\mathbf{A}^T$, involving both $2N(N-M)p$ operations at each step. When N is large, the computations involved prevent a real-time use of such techniques. Furthermore, even if real-time processing is not required, the signal typically has to be split into several chunks in order to perform the reconstruction. This can lead to artifacts in the borders of the chunk that compromise the quality of the approximation. In the following, we detail how these issues can be substantially alleviated.

2.2. CoSA: An accelerated ISTA algorithm

Let us consider the dictionary $\mathbf{A}$ of Eq. (??), which is composed of p translated waveforms and is widely used in many signal processing applications. The key point is realizing that the columns of $\mathbf{A}$ can be rearranged not according to the time blocks, but rather according to the shapes. That is, we define a *shape-block* matrix $\mathbf{B}_s = [\Gamma_s(t_i - t_k)]_{0 \leq i \leq N-1, 0 \leq k \leq N-M-1}$, and a shape based $N \times (N-M)p$ new dictionary as $\mathbf{B} = [\mathbf{B}_1 \ \mathbf{B}_2 \ \cdots \ \mathbf{B}_p]$. Figure 2 provides

a graphical explanation of the columns permutation to obtain the equivalent dictionary $\mathbf{B}$ from $\mathbf{A}$. Neglecting the influence of δ , Eq. (??) can therefore be rewritten as

$$\mathbf{y} = \mathbf{B}\tilde{\boldsymbol{\beta}} + \boldsymbol{\varepsilon} = \sum_{s=1}^p \mathbf{B}_s \tilde{\boldsymbol{\beta}}_s + \boldsymbol{\varepsilon},$$

where $\tilde{\boldsymbol{\beta}}$ is the vector obtained by permuting the entries of $\boldsymbol{\beta}$ according to the permutations of the columns performed to obtain $\mathbf{B}$ from $\mathbf{A}$; and $\tilde{\boldsymbol{\beta}}_s$ is the subvector of $\tilde{\boldsymbol{\beta}}$ related to $\mathbf{B}_s$, with $N - M$ entries.

Note that this model is equivalent to the previous one, i.e., applying (??) to (??) enables us to find the same estimate of $\boldsymbol{\beta}$ (after performing the inverse permutation to the entries of the ISTA solution) than applying (??) to (??) directly. Furthermore, using the fact that the shapes used in $\mathbf{B}$ have a finite length, the product $\mathbf{B}_s \tilde{\boldsymbol{\beta}}_s$ can be seen as the first N terms of the discrete-time convolution between $\boldsymbol{\Gamma}_s = [\Gamma_s(t_0) \dots \Gamma_s(t_{M-1})]$ and $\tilde{\boldsymbol{\beta}}_s$. It is well known that this convolution can be computed efficiently using the Fast Fourier Transform (FFT) procedure [17]. More precisely, if we denote by $\text{FFT}(\mathbf{x})$ and $\text{IFFT}(\mathbf{x})$ the N -Fast Fourier Transform and the N -inverse Fast Fourier Transform of the finite-length discrete signal $\mathbf{x}$, respectively, we have:

$$\boldsymbol{\Gamma}_s * \tilde{\boldsymbol{\beta}}_s = \text{IFFT}(\text{FFT}(\boldsymbol{\Gamma}_s) \times \text{FFT}(\tilde{\boldsymbol{\beta}}_s)) .$$

Consequently, the matrix-vector multiplication $\mathbf{A}\boldsymbol{\beta}^{(k)}$ appearing in (??) can be computed with an $O(pN \log_2 N)$ computational cost, which can be a significant improvement w.r.t. the $O(pN(N - M))$ cost of a conventional matrix-vector multiplication. This gain can be particularly significant if $N \gg M$, which is the case in numerous signal processing applications. Similarly, the multiplication by $\mathbf{B}^T$, $\mathbf{B}^T(\mathbf{y} - \mathbf{B}\tilde{\boldsymbol{\beta}}^{(k)})$, can be expressed, after permutations of the columns, as

$$\begin{bmatrix} \mathbf{B}_1^T \\ \mathbf{B}_2^T \\ \vdots \\ \mathbf{B}_p^T \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_p \end{bmatrix},$$

where the $r_s = \mathbf{y}_s - [\widetilde{\mathbf{B}}\widetilde{\boldsymbol{\beta}}^{(k)}]_s$ ($s = 1, \dots, p$) denote the residuals of the reconstruction at the k -th iteration, and each term in this matrix-vector product can be seen as the N last terms of the correlation between $\mathbf{\Gamma}_s$ and r_s , which can be computed with the same approach and complexity as above.

These considerations show that sparse reconstruction techniques with dictionaries based on translated waveforms can be computed in an efficient manner (since most computations can be parallelized accordingly to the number of waveforms), turning them into a very attractive choice for real-time or quasi-real-time applications. Furthermore, for off-line processing this approach allows us to work with much larger blocks than the standard ISTA method, thus minimizing border effects and increasing the precision of the reconstruction. Since many signal processing applications are based on this type of dictionaries and have to deal with large signals, CoSA can result in a substantial decrease in computational cost without a loss in performance, as shown in the numerical experiments of the following section.

3. Results

In this section, we provide simulation results illustrating the efficiency of the proposed approach. The simulations were performed on a PC with an i7, 3.4 GHz processor and 16 GB RAM. For the sake of reproducibility, the code used to obtain the following results is attached as supplementary material.

3.1. Description of the Experiments

In the first experiment, for a given signal size $N \geq 100$, $p = 20$ random waveforms were drawn from a multidimensional normal distribution, $\mathcal{N}(\mathbf{0}, 30\mathbf{I}_{30})$. The sparse signal is then simulated as follows: random starting points are generated using an homogeneous Poisson process with intensity 10, and for each starting point we select randomly a waveform from the previously generated ones, and a random amplitude uniformly in $[15, 20]$. An additional Gaussian noise with zero mean and standard deviation 0.01 is also added to the whole

signal. We then compute both the regular version of ISTA with a constant step size, as described in [3], and compare it with the proposed approach in terms of execution times. We let the size of the signal grow from $N = 100$ to $N = 30000$, and for each N , 100 runs of both ISTA and CoSA are performed. From the 100 executions times obtained, we compute an empirical 90% confidence interval from the fifth smallest to the fifth largest execution times, as well as the average execution time.

In order to illustrate the performance of CoSA on real datasets, as a second experiment we chose to measure its execution time on real-world spectroscopic data from CEA Saclay. In this experiment, $N = 100000$, thus forbidding the use of a standard implementation of ISTA. We used for the sparse reconstruction 20 Gamma waveforms, as described in [5].

3.2. Results and Discussion

Results on the execution times of ISTA and CoSA for the first example are displayed in Figure 3. Not surprisingly, both algorithms behave similarly for small values of N , with execution times ranging from 0.1 to 1 seconds. Indeed, when $N \leq 400$, it is more interesting to use regular ISTA, due to Matlab's capabilities regarding fast matrix-vector multiplications. However, the gain in speed using CoSA dramatically increases when $N \geq 500$, and CoSA can be almost 100 times faster for large values of N , while providing almost identical results. The proposed method has little effect on accuracy, the only variability stemming from the FFT operations. In practice, results obtained with CoSA are identical to the results obtained with ISTA, up to the machine precision, as can be seen in the supplementary material related to this paper. It is also noticeable that CoSA can operate on very large signals, when the full dictionary cannot be stored in the RAM of the computer used. Indeed, it can be seen from Figure 3 that standard ISTA could not be performed for $N \geq 15000$, whereas CoSA was still functioning. This fact makes the proposed approach relevant to deal with very large signals, as encountered in the fields of nuclear spectroscopy and biomedical signal processing, among others.

Results on the real spectroscopic data are presented in Figure 4. The execution time for $N = 100000$ was 21 seconds. We observe that the sparse reconstruction of the signal enables us to count the number of electrical pulses, after a thresholding operation such as the one described in [5]. In this framework, the proposed method allows us to process large portions of the signal that cannot be processed directly with standard ISTA. Consequently, the estimate of the activity obtained is more precise, since splitting the data into smaller parts would increase border effects and decrease the precision of the activity estimation.

4. Conclusion

In this contribution, we have derived a novel algorithm (CoSA) that enables us to perform sparse reconstruction quickly and efficiently for dictionaries based on translated waveforms, which are widely used in many signal processing applications. The performance of the proposed method was investigated through numerical simulations, which showed that the gain in time can be significant for very long signals, thus rendering CoSA attractive for real-time or quasi-real-time applications. Furthermore, taking into account that usually $N \gg M$ in real-world applications, this complexity can be reduced even more by using standard DSP techniques for the computation of fast convolutions, such as the overlap-add and overlap-save methods. These and other improvements will be explored in future contributions.

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$$\mathbf{A}_k = \left[\begin{array}{c} \mathbf{0}_{k-1 \times p} \\ \Gamma_s(t_i - t_k) \\ \mathbf{0}_{N-M-k+1 \times p} \end{array} \right]_{\substack{k \leq i \leq k+M-1 \\ 1 \leq s \leq p}} = \left[\begin{array}{c} \mathbf{0}_{k-1 \times p} \\ \text{5 wavy lines} \\ \mathbf{0}_{N-M-k+1 \times p} \end{array} \right]$$

} $k-1$
} M
} $N-M-k+1$

} p

Figure 1: Mathematical definition of $\mathbf{A}_k$ and graphical explanation for $s = 5$ candidates shapes.

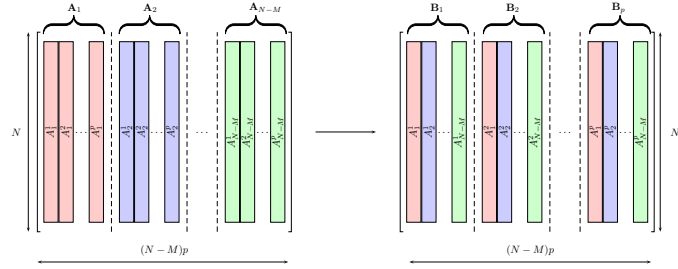


Figure 2: Graphical explanation of the columns permutation of the dictionary $\mathbf{A}$ (left) to obtain the equivalent dictionary $\mathbf{B}$ (right).

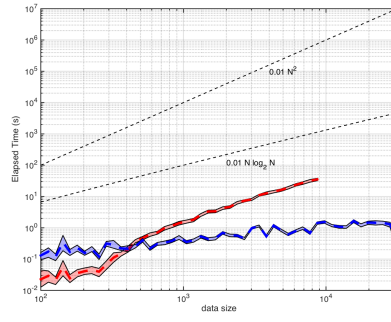


Figure 3: Comparison of the execution times between ISTA (red) and convISTA (blue), with 90% confidence intervals of the execution times.

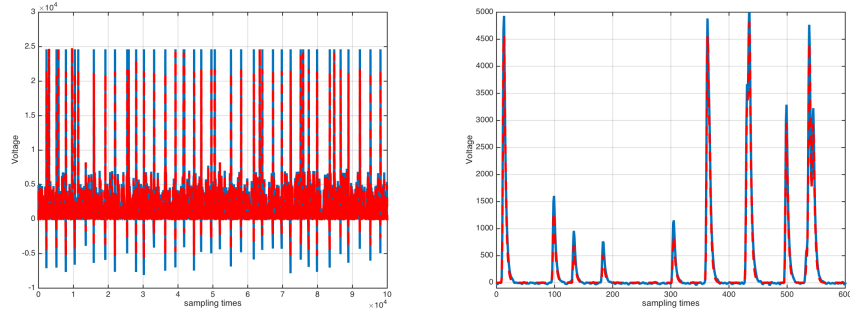


Figure 4: Results obtained on a sparse reconstruction of spectroscopic data using CoSA for $N = 100000$. The original signal is displayed in blue, its reconstruction in dotted red. Left: overall signal. Right: zoom on the first 600 sampling points.